

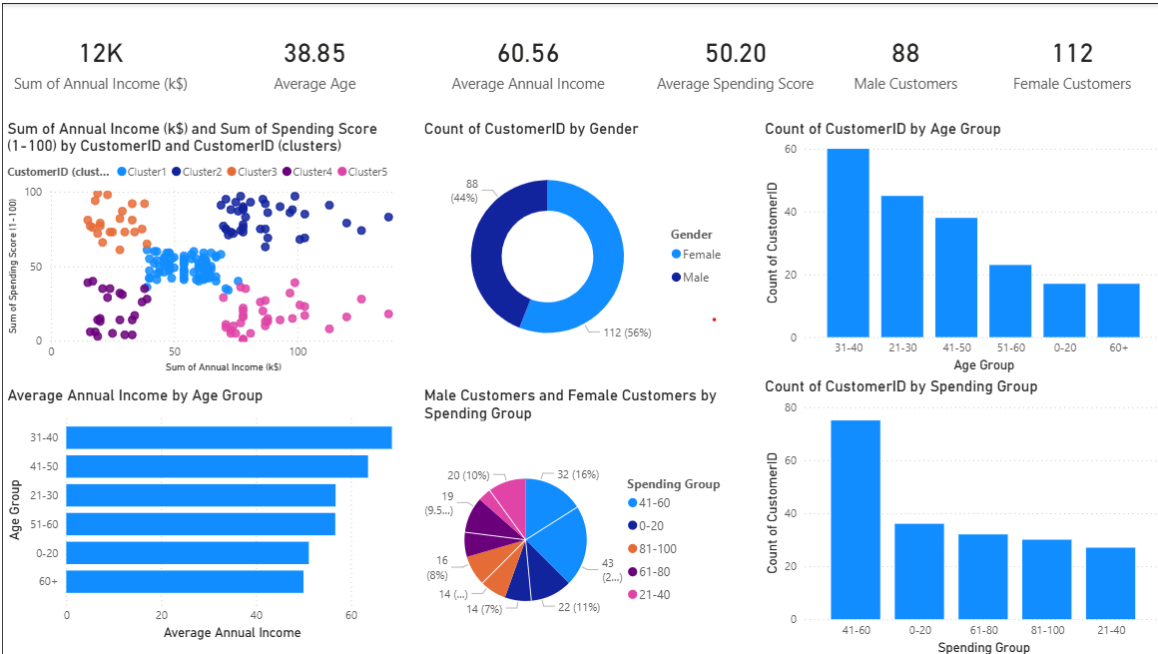
Customer Segmentation Analysis

Power BI Dashboard Report — Insights & Recommendations

Prepared by: Sumit Pawar

B.Tech (Artificial Intelligence & Data Science)

Tools Used: Power BI | Python (scikit-learn)



1. Executive Summary

This report presents an analysis of a customer segmentation dashboard built in Power BI, based on a dataset of 200 customers described by Customer ID, Gender, Age, Annual Income, and Spending Score (1-100). K-means style clustering was used to group customers into five behavioral segments, which are visualized alongside supporting demographic and spending breakdowns. The objective is to translate the dashboard's visuals into clear findings and actionable recommendations that a marketing or business team could use to design targeted strategies for each customer group.

The customer base skews slightly female (56% vs. 44% male), has an average age of 38.85 years, an average annual income of 60.56k\$, and an average spending score of 50.20. The most valuable and most vulnerable segments are the high-income/high-spending group and the low-income/high-spending group respectively — both are prime candidates for differentiated retention and financial-wellness strategies discussed in Section 6.

2. Dashboard Overview — Key Metrics

The dashboard's KPI cards summarize the overall customer base as follows:

Metric	Value
Sum of Annual Income	12K (k\$)
Average Age	38.85 years
Average Annual Income	60.56 k\$
Average Spending Score	50.20 / 100
Male Customers	88 (44%)
Female Customers	112 (56%)
Total Customers	200

3. Methodology

- **Data Preparation:** Customer demographic and transactional attributes (Age, Annual Income, Spending Score, Gender) were cleaned and loaded into Power BI for visualization; the underlying clustering was generated using an unsupervised learning approach (K-means, consistent with a scikit-learn workflow) on Income and Spending Score.
- **Clustering:** Customers were partitioned into 5 clusters based on Annual Income and Spending Score, chosen because this pair most clearly separates purchasing behavior from earning capacity.
- **Visualization:** Power BI was used to build an interactive dashboard with KPI cards, a cluster scatter plot, demographic breakdowns (gender, age group), and spending-group breakdowns (by count and by gender).
- **Interpretation:** Each cluster was profiled by its income and spending characteristics and translated into a business-friendly persona (Section 4).

4. Cluster Analysis

The scatter plot of Sum of Annual Income (k\$) vs. Sum of Spending Score (1-100) reveals five visually distinct clusters. Each represents a different combination of earning capacity and spending behavior:

Cluster	Profile (Income vs. Spending Score)	Approx. Size	Business Label
Cluster 1 (Light Blue)	Mid income (~40-70k\$), mid spending score (~40-60)	Largest segment	Average / Standard Customers
Cluster 2 (Navy Blue)	High income (~70-140k\$), low-to-mid spending score (~40-60, several under 60)	Medium	High Earners, Cautious Spenders
Cluster 3 (Orange)	Low-to-mid income (~15-40k\$), high spending score (~60-100)	Medium	Impulsive / Aspirational Spenders
Cluster 4 (Purple)	Low income (~15-40k\$), low spending score (~0-40)	Medium	Budget-Constrained Customers
Cluster 5 (Pink)	High income (~70-140k\$), high spending score (~60-100)	Smallest segment	Premium / High-Value Customers

The clearest business opportunities sit at the two extremes: Cluster 5 (high income, high spending) represents the most profitable customers, while Cluster 3 (low income, high spending) represents customers who may be over-extending relative to their income — a segment worth engaging carefully with budgeting tools or flexible payment options rather than aggressive upselling.

5. Demographic & Spending Insights

5.1 Gender Distribution

- Female customers make up 56% (112) of the base, while male customers make up 44% (88) — a modest but clear female majority.
- Marketing creative and channel mix should reflect this skew, without neglecting the sizeable male segment.

5.2 Age Distribution

- The 31-40 age group is the largest by count (~58 customers), followed by 21-30 (~48) and 41-50 (~38).
- 51-60, 0-20, and 60+ are comparatively small groups (roughly 17-22 customers each), suggesting the business's core base is working-age adults (21-50).
- Average Annual Income by age group follows a similar shape: 31-40 and 41-50 earn the most on average, while 0-20 and 60+ earn the least — consistent with typical career earning curves.

5.3 Spending Group Distribution

- The 41-60 spending-score band is by far the largest group (~75 customers), representing the 'average spender' majority.
- The remaining customers are fairly evenly spread across 0-20, 61-80, 81-100, and 21-40, each in the high-20s to high-30s range.
- Within the male/female breakdown by spending group, the 41-60 band again dominates for both genders, while the extreme bands (0-20 and 81-100) are smaller and more evenly split — indicating spending propensity is not strongly gender-driven in this dataset.

6. Key Insights

- **Balanced but female-leaning base:** A 56/44 female-to-male split means campaigns should be broadly appealing but can lean slightly toward channels and messaging that resonate with female shoppers.
- **Core demographic is 21-50:** Over 70% of customers fall in the 21-50 age range, and this group also earns the most — making it the highest-priority segment for revenue-driving campaigns.

- **Spending is concentrated in the 'average' band:** Most customers (41-60 spending score) are neither highly loyal big spenders nor disengaged — they represent the biggest growth opportunity if nudged toward higher engagement.
- **Income and spending are not always aligned:** Clusters 2 and 3 show that high income does not guarantee high spending, and low income does not guarantee low spending — a purely income-based targeting strategy would miss real behavioral differences.
- **A small, high-value segment exists:** Cluster 5 (high income, high spending) is the smallest but most profitable group and should receive VIP-level retention treatment.

7. Recommendations

7.1 Segment-Specific Strategies

- **Cluster 5 — Premium/High-Value:** Launch a loyalty or VIP tier with early access, exclusive offers, and personalized outreach to protect and grow this high-margin segment.
- **Cluster 2 — High Earners, Cautious Spenders:** Use trust-building content (reviews, guarantees, premium positioning) and highlight value/quality rather than discounts to convert latent spending power into purchases.
- **Cluster 3 — Aspirational Spenders:** Offer installment plans, loyalty points, or bundle deals that reward continued spending while being mindful of over-extension; avoid predatory upselling.
- **Cluster 4 — Budget-Constrained:** Focus on value-tier products, essential bundles, and price-sensitive promotions (e.g., seasonal discounts) rather than premium marketing.
- **Cluster 1 — Average Customers:** The largest segment; A/B test upsell and cross-sell offers here to shift more customers toward Cluster 5 behavior over time.

7.2 Cross-Cutting Recommendations

- Prioritize marketing spend on the 21-50 age range, which combines the largest population with the highest average income.
- Design a mid-tier loyalty program aimed at the 41-60 spending-score majority to nudge them toward higher engagement (61-80+).
- Investigate the 0-20 and 60+ age groups separately — their small size may reflect either low market penetration or a genuine niche audience worth a dedicated (lower-cost) campaign.
- Track cluster migration over time (e.g., quarterly) to measure whether campaigns are successfully moving customers from Cluster 1 toward Cluster 5.
- Extend the underlying dataset with purchase category, frequency, and channel data to move from demographic clustering to true behavioral/RFM-based segmentation.

8. Conclusion

The Power BI dashboard confirms that this customer base can be meaningfully split into five actionable segments using just income and spending score. While the largest opportunity lies in the broad, average-spending core (Clusters 1 and the 41-60 spending band), the greatest strategic value lies in protecting the small high-income/high-spending segment and carefully nurturing the low-income/high-spending segment. Applying the segment-specific recommendations above should improve marketing efficiency, customer retention, and overall revenue per customer.

This project demonstrates end-to-end customer analytics skills: data preparation, unsupervised clustering (Python/scikit-learn), interactive dashboard design (Power BI), and translation of visual analytics into business recommendations.